Scientists Confirm a Third Type of Magnetism

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For almost 90 years, scientists put magnetic materials into two groups. Now they have strong proof of a third group. A team led by SLAC found the proof, and the study was published in the journal Science.

The team studied a material called **manganese telluride**, or MnTe. They found a twist in its magnetic structure. The twist is called **chirality**, and it is like the shape of a spiral staircase. That twist is a key sign of the third kind of magnetism, called **altermagnetism**. Lead author Daniel Jost of SLAC said the results give strong proof that this third kind is real.

Magnetism comes from a property of electrons called spin. In **ferromagnets**, the spins point the same way and make the field of an ordinary magnet. In **antiferromagnets**, found in 1933, neighboring spins point opposite ways, so their effects cancel out. In altermagnets, the spins also alternate, so they show no overall magnetism. Yet they can carry spin currents, like **ferromagnets**. The discovery could lead to better memory systems, quantum devices, and data centers that use less energy.

Word Treasure

altermagnetism -- A third kind of magnetism where spins alternate but can still carry spin currents.

chirality -- A twist in a structure, like the shape of a spiral staircase.

manganese telluride -- The material the scientists studied to find proof of the third kind of magnetism.

ferromagnets -- Materials whose electron spins all point the same way and make an ordinary magnet.

antiferromagnets -- Materials whose neighboring spins point opposite ways, so their effects cancel out.

Background you need

Magnetism comes from tiny parts inside materials called electrons, and the way their spins point decides what kind of magnet a material is.

In some magnets the spins all point the same way and make a strong magnetic field, while in others the spins point in opposite ways and cancel each other out.

Altermagnets are a third kind where the spins alternate in a twisted pattern, so they show no overall magnetism but can still carry spin currents.

Break it down

WHO: A team led by SLAC, with lead author Daniel Jost

WHAT: Found strong proof of a third kind of magnetism, called altermagnetism, in a material called manganese telluride

WHERE: At SLAC, where the team studied manganese telluride

WHY: The twist called chirality in manganese telluride is a key sign of altermagnetism, which could lead to better memory systems, quantum devices, and data centers that use less energy

Why it matters

This discovery could one day make the computers, phones, and game consoles you use faster and use less energy.

Test what you remember

1. For almost 90 years, how many groups did scientists put magnetic materials into?

- (A) Three
- (B) Five
- (C) One
- (D) Two

2. What material did the team study to find proof of the third kind of magnetism?

- (A) Iron filings
- (B) Copper wire
- (C) Glass
- (D) Manganese telluride (MnTe)

3. What is the twist in the magnetic structure called?

- (A) Gravity
- (B) Electricity
- (C) Chirality
- (D) Magnetism

4. What does the article compare chirality to?

- (A) The shape of a spiral staircase
- (B) A round ball
- (C) A straight ruler
- (D) A flat sheet of paper

5. In ferromagnets, how do the spins of electrons point?

- (A) Up and down at random
- (B) The same way
- (C) In opposite ways
- (D) In circles

6. What is one thing the discovery could lead to?

- (A) Better memory systems
- (B) Brighter light bulbs
- (C) Faster bicycles
- (D) Stronger fridge magnets

Answer key (try the quiz first!): 1: D 2: D 3: C 4: A 5: B 6: A

Questions to think about

1. Lead author Daniel Jost of SLAC: He said the results give strong proof that the third kind of magnetism, altermagnetism, is real.

MY THOUGHTS (at least 20 words):